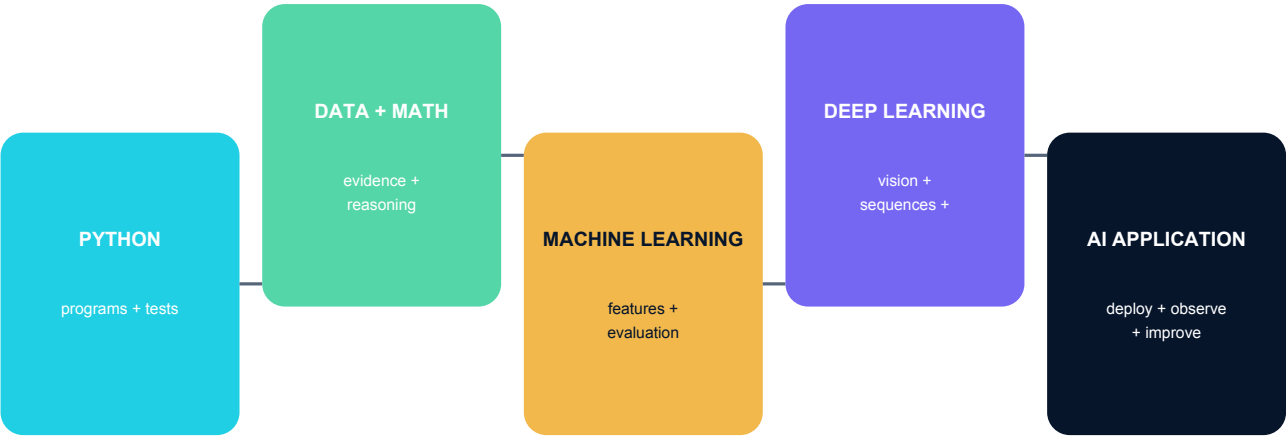


PROJECT WORKBOOK

Fifteen portfolio-ready briefs for planning, building, testing, documenting, and presenting the weekly projects.



15 WEEKS	125 LECTURES	FULL STACK
15 project briefs	Professional resource edition	Learn + build + explain

EVIDENCE OVER ACTIVITY

A notebook that runs is only a starting point. A strong portfolio artifact states the problem, preserves its inputs, tests failure behavior, reports honest metrics, and explains the tradeoffs another engineer must understand.

Minimum evidence bundle

- Problem and success: user, task, scope, baseline, metric, threshold, and risk.
- Reproduction: environment, dependencies, data source, seed, configuration, and one clean run command.
- Validation: split strategy, quality metrics, segment results, failure cases, and limitations.
- Engineering: modules or pipeline, tests, artifact version, interface contract, and security considerations.
- Operations: latency, resource or cost evidence, logs or monitoring, and rollback or recovery where relevant.
- Communication: README, architecture or workflow visual, result table, demo path, and next experiment.

Pythonic Code and Project Work

Deliverable. A command-line Python utility with modules, tests, safe file I/O, and a concise README.

PROBLEM	Who needs this result, and which decision or action will it support?
SUCCESS	Write a baseline, primary metric, acceptable threshold, and one risk guardrail.
INPUTS	List data, code, model or provider, versions, schema, credentials, and constraints.
PLAN	Write 4-7 ordered milestones, including a small end-to-end proof and final reproduction step.

EVIDENCE CHECKLIST

- ☐ The project rebuilds from a clean environment with a single documented command.
- ☐ Run from a clean virtual environment; test normal, boundary, and invalid inputs; use readable names and small functions.
- ☐ A simple baseline and at least one failure or invalid-input case are documented.
- ☐ Metrics include uncertainty or segment analysis where it can change the conclusion.
- ☐ Secrets, sensitive data, licensing, and model/provider limitations are addressed.
- ☐ README includes scope, setup, architecture, results, limitations, and next experiment.

RESULTS

Metric / artifact	Baseline	Result	Decision

RETROSPECTIVE

What changed your conclusion?	
Where does the system fail?	
What would production require?	
What is the next experiment?	

Exploratory Data Analysis (EDA) Project

Deliverable. An EDA report with a data dictionary, cleaning log, reproducible plots, findings, caveats, and next questions.

PROBLEM	Who needs this result, and which decision or action will it support?
SUCCESS	Write a baseline, primary metric, acceptable threshold, and one risk guardrail.
INPUTS	List data, code, model or provider, versions, schema, credentials, and constraints.
PLAN	Write 4-7 ordered milestones, including a small end-to-end proof and final reproduction step.

EVIDENCE CHECKLIST

- ☐ The project rebuilds from a clean environment with a single documented command.
- ☐ State inputs, assumptions, output contract, metric, failure case, version information, and the next decision.
- ☐ A simple baseline and at least one failure or invalid-input case are documented.
- ☐ Metrics include uncertainty or segment analysis where it can change the conclusion.
- ☐ Secrets, sensitive data, licensing, and model/provider limitations are addressed.
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Math-Driven Mini Project – Linear Regression from Scratch

Deliverable. Linear regression from scratch with matrix operations, a numerical gradient check, loss curve, and comparison to a library baseline.

PROBLEM	Who needs this result, and which decision or action will it support?
SUCCESS	Write a baseline, primary metric, acceptable threshold, and one risk guardrail.
INPUTS	List data, code, model or provider, versions, schema, credentials, and constraints.
PLAN	Write 4-7 ordered milestones, including a small end-to-end proof and final reproduction step.

EVIDENCE CHECKLIST

- ☐ The project rebuilds from a clean environment with a single documented command.
- ☐ State inputs, assumptions, output contract, metric, failure case, version information, and the next decision.
- ☐ A simple baseline and at least one failure or invalid-input case are documented.
- ☐ Metrics include uncertainty or segment analysis where it can change the conclusion.
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Statistical Analysis Project – Analyzing Real-World Data

Deliverable. A statistical decision memo with sampling assumptions, confidence interval, hypothesis test, effect size, and limitations.

PROBLEM	Who needs this result, and which decision or action will it support?
SUCCESS	Write a baseline, primary metric, acceptable threshold, and one risk guardrail.
INPUTS	List data, code, model or provider, versions, schema, credentials, and constraints.
PLAN	Write 4-7 ordered milestones, including a small end-to-end proof and final reproduction step.

EVIDENCE CHECKLIST

- ☐ The project rebuilds from a clean environment with a single documented command.
- ☐ Report sample definition, assumptions, interval or effect size, multiplicity, practical significance, and limits on causal claims.
- ☐ A simple baseline and at least one failure or invalid-input case are documented.
- ☐ Metrics include uncertainty or segment analysis where it can change the conclusion.
- ☐ Secrets, sensitive data, licensing, and model/provider limitations are addressed.
- ☐ README includes scope, setup, architecture, results, limitations, and next experiment.

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Supervised Learning Mini Project

Deliverable. A supervised-learning baseline with leakage-safe splits, cross-validation, evaluation by segment, and an error-analysis report.

PROBLEM	Who needs this result, and which decision or action will it support?
SUCCESS	Write a baseline, primary metric, acceptable threshold, and one risk guardrail.
INPUTS	List data, code, model or provider, versions, schema, credentials, and constraints.
PLAN	Write 4-7 ordered milestones, including a small end-to-end proof and final reproduction step.

EVIDENCE CHECKLIST

- ☐ The project rebuilds from a clean environment with a single documented command.
- ☐ State inputs, assumptions, output contract, metric, failure case, version information, and the next decision.
- ☐ A simple baseline and at least one failure or invalid-input case are documented.
- ☐ Metrics include uncertainty or segment analysis where it can change the conclusion.
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Cross-Validation and Hyperparameter Tuning

Deliverable. A reusable preprocessing-and-model pipeline that handles missing, numeric, and categorical inputs without train/serve skew.

PROBLEM	Who needs this result, and which decision or action will it support?
SUCCESS	Write a baseline, primary metric, acceptable threshold, and one risk guardrail.
INPUTS	List data, code, model or provider, versions, schema, credentials, and constraints.
PLAN	Write 4-7 ordered milestones, including a small end-to-end proof and final reproduction step.

EVIDENCE CHECKLIST

- ☐ The project rebuilds from a clean environment with a single documented command.
- ☐ Fit preprocessing on training folds only; test unseen categories, missing values, feature order, and schema at prediction time.
- ☐ A simple baseline and at least one failure or invalid-input case are documented.
- ☐ Metrics include uncertainty or segment analysis where it can change the conclusion.
- ☐ Secrets, sensitive data, licensing, and model/provider limitations are addressed.
- ☐ README includes scope, setup, architecture, results, limitations, and next experiment.

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Ensemble Learning Project – Comparing Models on a Real Dataset

Deliverable. An ensemble benchmark comparing a linear baseline, random forest, and boosting model on quality, calibration, latency, and size.

PROBLEM	Who needs this result, and which decision or action will it support?
SUCCESS	Write a baseline, primary metric, acceptable threshold, and one risk guardrail.
INPUTS	List data, code, model or provider, versions, schema, credentials, and constraints.
PLAN	Write 4-7 ordered milestones, including a small end-to-end proof and final reproduction step.

EVIDENCE CHECKLIST

- ☐ The project rebuilds from a clean environment with a single documented command.
- ☐ State inputs, assumptions, output contract, metric, failure case, version information, and the next decision.
- ☐ A simple baseline and at least one failure or invalid-input case are documented.
- ☐ Metrics include uncertainty or segment analysis where it can change the conclusion.
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Optimization Project – Building and Tuning a Final Model

Deliverable. A tuning study with an explicit search space, budget, trial log, validation uncertainty, untouched test result, and compute cost.

PROBLEM	Who needs this result, and which decision or action will it support?
SUCCESS	Write a baseline, primary metric, acceptable threshold, and one risk guardrail.
INPUTS	List data, code, model or provider, versions, schema, credentials, and constraints.
PLAN	Write 4-7 ordered milestones, including a small end-to-end proof and final reproduction step.

EVIDENCE CHECKLIST

- ☐ The project rebuilds from a clean environment with a single documented command.
- ☐ Record distributions, folds, seed, trials, failures, runtime, best score uncertainty, and untouched final test performance.
- ☐ A simple baseline and at least one failure or invalid-input case are documented.
- ☐ Metrics include uncertainty or segment analysis where it can change the conclusion.
- ☐ Secrets, sensitive data, licensing, and model/provider limitations are addressed.
- ☐ README includes scope, setup, architecture, results, limitations, and next experiment.

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Neural Network Project – Image Classification on CIFAR-10

Deliverable. A CIFAR-10 classifier with reproducible training, learning curves, confusion matrix, failure gallery, and saved best checkpoint.

PROBLEM	Who needs this result, and which decision or action will it support?
SUCCESS	Write a baseline, primary metric, acceptable threshold, and one risk guardrail.
INPUTS	List data, code, model or provider, versions, schema, credentials, and constraints.
PLAN	Write 4-7 ordered milestones, including a small end-to-end proof and final reproduction step.

EVIDENCE CHECKLIST

- ☐ The project rebuilds from a clean environment with a single documented command.
- ☐ Assert tensor shapes, overfit a tiny batch, inspect loss and gradients, separate train/eval behavior, and save the best checkpoint.
- ☐ A simple baseline and at least one failure or invalid-input case are documented.
- ☐ Metrics include uncertainty or segment analysis where it can change the conclusion.
- ☐ Secrets, sensitive data, licensing, and model/provider limitations are addressed.
- ☐ README includes scope, setup, architecture, results, limitations, and next experiment.

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CNN Project – Image Classification on Fashion MNIST or CIFAR-10

Deliverable. A CNN classifier with augmentation ablation, tensor-shape trace, per-class metrics, and inference benchmark.

PROBLEM	Who needs this result, and which decision or action will it support?
SUCCESS	Write a baseline, primary metric, acceptable threshold, and one risk guardrail.
INPUTS	List data, code, model or provider, versions, schema, credentials, and constraints.
PLAN	Write 4-7 ordered milestones, including a small end-to-end proof and final reproduction step.

EVIDENCE CHECKLIST

- ☐ The project rebuilds from a clean environment with a single documented command.
- ☐ Trace tensor shapes and receptive field; visualize errors by class; report augmentation, normalization, latency, and parameter count.
- ☐ A simple baseline and at least one failure or invalid-input case are documented.
- ☐ Metrics include uncertainty or segment analysis where it can change the conclusion.
- ☐ Secrets, sensitive data, licensing, and model/provider limitations are addressed.
- ☐ README includes scope, setup, architecture, results, limitations, and next experiment.

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RNN Project – Text Generation or Sentiment Analysis

Deliverable. A sentiment or text-generation sequence model with preprocessing contract, padding policy, baseline, and qualitative error analysis.

PROBLEM	Who needs this result, and which decision or action will it support?
SUCCESS	Write a baseline, primary metric, acceptable threshold, and one risk guardrail.
INPUTS	List data, code, model or provider, versions, schema, credentials, and constraints.
PLAN	Write 4-7 ordered milestones, including a small end-to-end proof and final reproduction step.

EVIDENCE CHECKLIST

- ☐ The project rebuilds from a clean environment with a single documented command.
- ☐ Verify sequence order, mask and padding, truncation length, hidden-state reset, gradient norm, and evaluation on long examples.
- ☐ A simple baseline and at least one failure or invalid-input case are documented.
- ☐ Metrics include uncertainty or segment analysis where it can change the conclusion.
- ☐ Secrets, sensitive data, licensing, and model/provider limitations are addressed.
- ☐ README includes scope, setup, architecture, results, limitations, and next experiment.

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RETROSPECTIVE

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Transformer Project – Text Summarization or Translation

Deliverable. A transformer summarization or translation application with baseline comparison, evaluation rubric, latency, and failure cases.

PROBLEM	Who needs this result, and which decision or action will it support?
SUCCESS	Write a baseline, primary metric, acceptable threshold, and one risk guardrail.
INPUTS	List data, code, model or provider, versions, schema, credentials, and constraints.
PLAN	Write 4-7 ordered milestones, including a small end-to-end proof and final reproduction step.

EVIDENCE CHECKLIST

- ☐ The project rebuilds from a clean environment with a single documented command.
- ☐ Track tokenization, mask semantics, sequence length, hidden shapes, task metric, latency, memory, and failure cases.
- ☐ A simple baseline and at least one failure or invalid-input case are documented.
- ☐ Metrics include uncertainty or segment analysis where it can change the conclusion.
- ☐ Secrets, sensitive data, licensing, and model/provider limitations are addressed.
- ☐ README includes scope, setup, architecture, results, limitations, and next experiment.

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Transfer Learning Project – Fine-Tuning for a Custom Task

Deliverable. A fine-tuned pretrained model with freeze/unfreeze schedule, domain-shift analysis, scratch comparison, and model card.

PROBLEM	Who needs this result, and which decision or action will it support?
SUCCESS	Write a baseline, primary metric, acceptable threshold, and one risk guardrail.
INPUTS	List data, code, model or provider, versions, schema, credentials, and constraints.
PLAN	Write 4-7 ordered milestones, including a small end-to-end proof and final reproduction step.

EVIDENCE CHECKLIST

- ☐ The project rebuilds from a clean environment with a single documented command.
- ☐ Record base checkpoint, frozen parameters, learning rates, unfreezing schedule, domain shift, forgetting, and comparison with scratch.
- ☐ A simple baseline and at least one failure or invalid-input case are documented.
- ☐ Metrics include uncertainty or segment analysis where it can change the conclusion.
- ☐ Secrets, sensitive data, licensing, and model/provider limitations are addressed.
- ☐ README includes scope, setup, architecture, results, limitations, and next experiment.

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MLOps Project – Build and Deploy a Model End-to-End

Deliverable. An end-to-end model service with versioned data/model, automated tests, packaged artifact, API schema, deployment, monitoring, and rollback note.

PROBLEM	Who needs this result, and which decision or action will it support?
SUCCESS	Write a baseline, primary metric, acceptable threshold, and one risk guardrail.
INPUTS	List data, code, model or provider, versions, schema, credentials, and constraints.
PLAN	Write 4-7 ordered milestones, including a small end-to-end proof and final reproduction step.

EVIDENCE CHECKLIST

- ☐ The project rebuilds from a clean environment with a single documented command.
- ☐ Capture code/data/model versions, dependency lock, schema, smoke and load tests, health checks, metrics, security, and rollback.
- ☐ A simple baseline and at least one failure or invalid-input case are documented.
- ☐ Metrics include uncertainty or segment analysis where it can change the conclusion.
- ☐ Secrets, sensitive data, licensing, and model/provider limitations are addressed.
- ☐ README includes scope, setup, architecture, results, limitations, and next experiment.

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Generative AI Project – Build Your Own Chatbot or Content Generator

Deliverable. A provider-abstracted chatbot or content tool with secure key handling, retrieval or tools, structured output, evaluations, safety checks, and cost report.

PROBLEM	Who needs this result, and which decision or action will it support?
SUCCESS	Write a baseline, primary metric, acceptable threshold, and one risk guardrail.
INPUTS	List data, code, model or provider, versions, schema, credentials, and constraints.
PLAN	Write 4-7 ordered milestones, including a small end-to-end proof and final reproduction step.

EVIDENCE CHECKLIST

- ☐ The project rebuilds from a clean environment with a single documented command.
- ☐ Keep keys server-side; pin provider/model metadata; add timeout, retry, rate limits, schema checks, logging, evaluations, and cost limits.
- ☐ A simple baseline and at least one failure or invalid-input case are documented.
- ☐ Metrics include uncertainty or segment analysis where it can change the conclusion.
- ☐ Secrets, sensitive data, licensing, and model/provider limitations are addressed.
- ☐ README includes scope, setup, architecture, results, limitations, and next experiment.

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